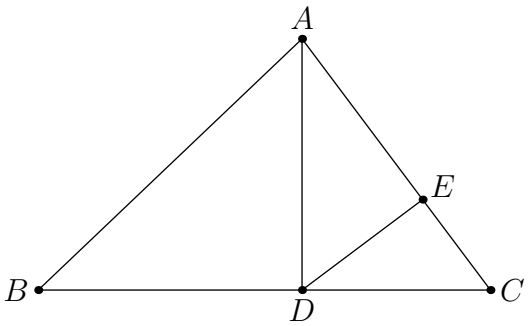
2024 SSMO Relay Round 1

SMO Team

RR 1 Part 1: Let AD be an altitude of triangle ABC and let DE be an altitude of triangle ACD. If $AB = 29$, $CE = 9$, and $DE = 12$, find the area of triangle ABC.



RR 1 Part 2: Let $T = TNYWR$. A circular necklace is called *interesting* if it has T black beads and T white beads. A move consists of cutting out a segment of consecutive beads and reattaching it in reverse. Find the least positive integer x such that any *interesting* necklace can be changed into one in which all T black beads are consecutive using at most x moves.

RR 1 Part 3: Let $T = TNYWR$. In a circle, there are T people. $T - 2$ of them have red shoes, and two of them have blue shoes. First, $\left\lfloor \frac{T}{2} \right\rfloor$ people with red shoes are eliminated one at a time, each chosen uniformly at random from the remaining people with red shoes. Then, people are eliminated one at a time, each chosen uniformly at random from everyone remaining, until there is only one person left. The probability that this person has blue shoes can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

2024 SSMO Relay Round 2

SMO Team

RR 2 Part 1: In a regular hexagon $ABCDEF$, let X be a point inside the hexagon such that $XA = XB = 3$. If the area of the hexagon is $6\sqrt{3}$, then XE^2 can be expressed as $a - b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a + b + c$.

RR 2 Part 2: Let $T = TNYWR$. If

$$a = \sum_{n=1}^{T} n(n+1)(n+2),$$

find the last three digits of a.

RR 2 Part 3: Let $T = TNYWR$. Let $A = (0, 0)$, $B = (0, T)$, $C = (T, T)$, and $D = (T, 0)$. Find the perimeter of the set S of all points P inside square $ABCD$ for which $AP + CP \geq BP + DP$.

2024 SSMO Relay Round 3

SMO Team

RR 3 Part 1: Alice and Bob are on the second floor of a building, right next to the top of an escalator moving upwards. Alice and Bob each run at 200 steps per minute relative to the ground or the escalator they are on, and each escalator is a total of 225 steps. Alice chooses to run down the escalator moving upwards, while Bob chooses to run to the escalator moving downwards, 300 steps away, and then run down that escalator. Both escalators move at the same speed. If Alice and Bob hit the bottom floor at the exact same time, find the speed of the escalators in terms of steps per minute.

RR 3 Part 2: Let $T = TNYWR$. Find the greatest odd integer n for which $n^2 + (T-1)n$ is a perfect square.

RR 3 Part 3: Let $T = TNYWR$. Riley and Boris are playing a game on a $(T-1) \times (T-1)$ grid of dots. The game alternates turns and starts with Riley. Each turn, a player draws a segment connecting two dots exactly 1 unit apart that are not already connected. The game ends when every such pair of dots is connected, and the player who draws the last segment loses. Given that Riley plays optimally and Boris plays randomly, the probability that Riley wins can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

2024 SSMO Relay Round 4

SMO Team

RR 4 Part 1: Freddy the Frog can jump 1 unit right or up. He is at $(0, 0)$ and wants to get to $(7, 4)$. However, he is scared of points $(3, 1)$ and $(3, 2)$ and will not hop onto those points. How many ways can he reach his destination?

RR 4 Part 2: Let $T = TNYWR$. Regular octagon $OLYMPIAD$ is inscribed in circle Q. Circle Q has area $T\pi$. The area of octagon $OLYMPIAD$ can be expressed as $a\sqrt{b}$, where a and b are positive integers and b is not divisible by the square of any prime. Find $a + b$.

RR 4 Part 3: Let $T = TNYWR$. Real numbers a, b, and c satisfy

$$\begin{aligned} a + b &= -c, \\ a^3 - abc &= 4, \text{ and} \\ b^3 - abc &= T. \end{aligned}$$

Find the value of $abc - c^3$.

2024 SSMO Relay Round 5

SMO Team

RR 5 Part 1: Let the super factorial $!(n)$ be defined on positive integers as $\prod_{i=1}^{n} i!$. Find the largest positive integer k such that there are exactly k positive integers n such that $!(n)$ has fewer than k trailing zeroes.

RR 5 Part 2: Let $T = TNYWR$. In the game of high and low, the computer chooses two integers uniformly at random without replacement from the set $\{1, 2, 3, \ldots, T\}$. The computer displays the first integer and asks the player if the second integer is higher or lower. Given that the player always plays optimally, the probability of guessing correctly can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

RR 5 Part 3: Let $T = TNYWR$. Let k be the largest prime factor of T. How many values of x satisfy

$$\sin(x^2) + \cos(x^2) = \sin^2 x + \cos^2 x \quad \text{and} \quad -k \leq x \leq k?$$